

- It supplies the stored information to the other units of the computer as and when required.
- It is less expensive and has higher storage capacity than the primary memory.

INPUT DEVICES : Following are few of the important input devices which are used in a computer:

- Keyboard
- Mouse
- Joy Stick
- Light pen
- Track Ball
- Scanner
- Graphic Tablet
- Microphone
- Magnetic Ink Card Reader MICR
- Optical Character Reader OCR
- Bar Code Reader
- Optical Mark Reader OMR

Keyboard :

- Keyboard is the most common and very popular input device which helps in inputting data to the computer.
- The layout of the keyboard is like that of traditional typewriter, although there are some additional keys provided for performing additional functions.
- Keyboards are of two sizes 84 keys or 101/102 keys, but now keyboards with 104 keys or 108 keys are also available for Windows and Internet.

The keys on the keyboard are as follows:

- **Typing Keys:** These keys include the letter keys A - Z and digit keys 0 - 9 which generally give same layout as that of typewriters.
- **Numeric Keypad:** It is used to enter numeric data or cursor movement. Generally, it consists of a set of, 17 keys that are laid out in the same configuration used by most adding machines and calculators.
- **Function Keys:** The twelve function keys are present on the keyboard which is arranged in a row at the top of the keyboard. Each function key has unique meaning and is used for some specific purpose.
- **Control keys:** These keys provide cursor and screen control. It includes four directional arrow keys. Control keys also include Home, End, Insert, Delete, Page Up, Page Down, Control Ctrl, Alternate Alt and Escape Esc.
- **Special Purpose Keys:** Keyboard also contains some special purpose keys such as Enter, Shift, Caps Lock, Num Lock, Space bar, Tab, and Print Screen.

Mouse :

- Mouse is most popular pointing device.
- It is a very famous cursor - control device having a small palm size box with a round ball at its base which senses the movement of mouse and sends corresponding signals to CPU when the mouse buttons are pressed.
- Generally it has two buttons called left and right button and a wheel is present between the buttons.
- Mouse can be used to control the position of cursor on screen, but it cannot be used to enter text into the computer.

Advantages

- Easy to use
- Not very expensive
- Moves the cursor faster than the arrow keys of keyboard.

Joystick :

- Joystick is also a pointing device which is used to move cursor position on a monitor screen.
- It is a stick having a spherical ball at its both lower and upper ends.
- The joystick can be moved in all four directions. The function of joystick is similar to that of a mouse.
- It is mainly used in Computer Aided Designing CAD and playing computer games.

Light Pen :

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Light Pen :

- Light pen is a pointing device which is similar to a pen.
- It is used to select a displayed menu item or draw pictures on the monitor screen.
- When the tip of a light pen is moved over the monitor screen and pen button is pressed, its photocell sensing element detects the screen location and sends the corresponding signal to the CPU.

Track Ball :

- Track ball is an input device that is mostly used in notebook or laptop computer, instead of a mouse.
- This is a ball which is half inserted and by moving fingers on ball, pointer can be moved.
- Since the whole device is not moved, a track ball requires less space than a mouse.
- A track ball comes in various shapes like a ball, a button and a square.

Scanner :

- Scanner is an input device which works more like a photocopy machine.
- It is used when some information is available on a paper and it is to be transferred to the hard disc of the computer for further manipulation.
- Scanner captures images from the source which are then converted into the digital form that can be stored on the disc.
- These images can be edited before they are printed.

Digitizer :

- Digitizer is an input device which converts analog information into digital form.
- Digitizer can convert a signal from the television or camera into a series of numbers that could be stored in a computer.
- Digitizer is also known as Tablet or Graphics Tablet because it converts graphics and pictorial data into binary inputs.
- A graphic tablet as digitizer is used for doing fine works of drawing and image manipulation applications.

Microphone :

- Microphone is an input device to input sound that is then stored in digital form.
- The microphone is used for various applications like adding sound to a multimedia presentation or for mixing music.

Magnetic Ink Card Reader (MICR) :

- MICR input device is generally used in banks because of a large number of cheques to be processed every day.
- The bank's code number and cheque number are printed on the cheques with a special type of ink that contains particles of magnetic material that are machine readable.
- This reading process is called Magnetic Ink Character Recognition MICR.
- The main advantages of MICR are that it is fast and less error prone.

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Optical Character Reader (OCR) :

- OCR is an input device used to read a printed text.
- OCR scans text optically character by character, converts them into a machine-readable code and stores the text on the system memory.

Bar Code Readers :

- Bar Code Reader is a device used for  which are in the form of light and dark lines.
- Bar coded data is generally used in labelling goods, numbering the books etc.
- It may be a hand held scanner or may be embedded in a stationary scanner.
- Bar Code Reader scans a bar code image, converts it into an alphanumeric value which is then fed to the computer to which bar code reader is connected.

Optical Mark Reader (OMR) :

- OMR is a special type of optical scanner used to recognize the type of mark made by pen or pencil.
- It is used where one out of a few alternatives is to be selected and marked.
- It is specially used for checking the answer sheets of examinations having multiple choice questions.

OUTPUT DEVICES :

Following are few of the important output devices which are used in a computer.

- ✓ Monitors
- ✓ Printers
- ✓ Headphones
- ✓ Speakers
- ✓ Projector
- ✓ Plotter

Monitors

- Monitors, commonly called as Visual Display Unit VDU, are the main output device of a computer.
- It forms images from tiny dots, called pixels that are arranged in a rectangular form.
- The sharpness of the image depends upon the number of pixels.
- There are two kinds of viewing screen used for monitors :
 - a) Cathode Ray Tube (CRT)

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 - b) Flat Panel Display

a) Cathode - Ray Tube (CRT) Monitor

- The CRT display is made up of small picture elements called pixels. The smaller the pixels, the better the image clarity, or resolution.
- A finite number of characters can be displayed on a screen at once.
- Most screens are capable of displaying 80 characters of data horizontally and 25 lines vertically.

There are some disadvantages of CRT:

Large in Size

High power consumption

111%

b) Flat - Panel Display Monitor

- The flat-panel display refers to a class of video - devices that have reduced volume, weight and power requirement in comparison to the CRT.
- Current uses of flat-panel displays include calculators, video games, monitors, laptop computer, graphics display.

The flat-panel display is divided into two categories:

Emissive Displays - The emissive displays are devices that convert electrical energy into light. Example are plasma panel and LED (Light Emitting Diodes).

Non-Emissive Displays - The Non-emissive displays use optical effects to convert sunlight or light from some other source into graphics patterns, Example is LCD (Liquid - Crystal Device).

Printers

- Printer is an output device, which is used to print information on paper.
- There are two types of printers:
 - Impact Printers
 - Non-Impact Printers

1) Impact Printers

- The impact printers print the characters by striking them on the ribbon which is then pressed on the paper.

Characteristics of Impact Printers :

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Characteristics of Impact Printers :

- Very low consumable costs
- Very noisy
- Useful for bulk printing due to low cost.

These printers are of two types

- Character printers
- Line printers

I. Character Printers

- Character printers are the printers which print one character at a time.
- These are further divided into two types:
 - Dot Matrix Printer (DMP)
 - Daisy Wheel

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i) Dot Matrix Printer

- In the market one of the most popular printers is Dot Matrix Printer.
- These printers are popular because of their ease of printing and economical price.
- Each character printed is in form of pattern of dots and head consists of a Matrix of Pins of size 5×7 , 7×9 , 9×7 or 9×9 which come out to form a character that is why it is called Dot Matrix Printer.

Advantages

- Inexpensive
- Widely Used
- Other language characters can be printed

Disadvantages

- Slow Speed
- Poor Quality

ii) Daisy Wheel

- Head is lying on a wheel and pins corresponding to characters are like petals of Daisy flower that is why it is called Daisy Wheel Printer.
- These printers are generally used for word-processing in offices which require a few letters to be sent here and there with very nice quality.

Advantages

- More reliable than DMP
- Better quality
- The fonts of character can be easily changed

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Disadvantages

- Slower than DMP Noisy
- More expensive than DMP.

II. Line Printers

- Line printers are the printers which print one line at a time.
- These are of further two types

a) Drum Printer b) Chain Printer

a) Drum Printer :

- This printer is like a drum in shape so it is called drum printer.
- Type of line printer that uses rotating drum to print text onto paper.
- In this printer, all the characters are not printed at one time but their printing speed is very high.
- One rotation of drum prints one line.
- Drum printers are fast in speed and can print 1000 lines per minute.

Advantages

- Very high speed

Disadvantages

- Very expensive
- Characters fonts cannot be changed.

b) Chain Printer :

- In this printer, chain of character sets is used so it is called Chain Printer.
- Chain printer are one of the fastest impact printers that can produce up to 400 to 2500 characters per second.

Advantages

- Character fonts can easily be changed.
- Different languages can be used with the same printer.

Disadvantages

- Noisy

2) Non - impact Printers :

- Non-impact printers print the characters without using ribbon.
- These printers print a complete page at a time so they are also called as Page Printers.
- These printers are of two types

i) Laser Printers ii) Inkjet Printers

Characteristics of Non-impact Printers :

- Faster than the impact printers.
- They are not noisy.
- High quality

- Characters fonts cannot be changed.

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Characteristics of Non-impact Printers :

- Faster than the impact printers.
- They are not noisy.
- High quality
- Support many fonts and different character size.

i) Laser Printers

- These are non-impact page printers.
- They use laser lights to produce the dots needed to form the characters to be printed on a page.

Advantages

- Very high speed
- Very high quality output
- Give good graphics quality
- Support many fonts and different character size.

Disadvantages

- Expensive.
- Cannot be used to produce multiple copies of a document in a single printing.

ii) Inkjet Printers

- Inkjet printers are non-impact character printers based on a relatively new technology.
- They print characters by spraying small drops of ink onto paper.
- Inkjet printers produce high quality output with presentable features.
- Colour printing is also possible.
- Some models of Inkjet printers can produce multiple copies of printing also.

Advantages

- High quality printing.
- More reliable.

Disadvantages

- Expensive as cost per page is high.
- Slow as compared to laser printer.

Plotter Printers

- Plotter is an output device that draws pictures on paper based on commands from a computer.

Disadvantages

- Expensive.
- Cannot be used to produce multiple copies of a document in a single printing.

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Plotter Printers

- Plotter is an output device that draws pictures on paper based on commands from a computer.
- Plotters differ from printers in that they draw lines using a pen. As a result, they can produce continuous lines, whereas printers can only simulate lines by printing a closely spaced series of dots.
- Multicolor plotters use different-colored pens to draw different colors.
- In general, plotters are considerably more expensive than printers.
- They are used in engineering applications where precision is mandatory.

Speakers :

- The speakers are the output units.
- The sound signals from analog/digital are converted into audible frequency in the speakers and produce voice output (audio data).
- Using speakers along with speech synthesizer software, the computer can provide voice output.
- Voice output has become very common in many places like airlines, banks, automatic telephone enquiry system etc.
- Users can also hear music/ songs using the voice output system.
- The latest speakers are crystal clear audio. Walkman speakers are smaller in size.

Storage devices :

- These are essential hardware components that store digital data, allowing users to save, retrieve, and manage information.
- They vary in speed, capacity, and permanence, catering to different storage needs.

Types of Storage Devices:

1. Hard Disk Drive (HDD) :

- A traditional storage device that uses magnetic disks to read and write data.

Types:

- ❖ Internal HDD : Installed inside the computer, typically used for primary storage.
- ❖ External HDD : Portable drives that connect via USB or other interfaces for additional storage.